

CV Maker — Project Overview

Project: CV Maker

Technology: Python/FastAPI backend, React/Vite + Svelte frontends, PostgreSQL, SQLAlchemy, Playwright scraper

Location: `/home/dominic/Documents/CV MAKER`

Report #: 1/20

Overview

Full-stack CV builder with job scraping, two frontends (React/Vite + Svelte), and PDF generation.

Purpose

Builds professional CVs from scraped job data, with dual frontend support.

Status

Active development. See report 16 (Known Issues) and report 19 (Improvements).

Key Features

• Job scraping from remote OK • CV generation with multiple templates • Dual frontends: React/Vite + Svelte • PDF export • User authentication

Entry Points

Backend: `main.py`; Frontend Svelte: `cd svelte-frontend && npm run dev`; Frontend React: `cd frontend && npm run dev`

Detailed Architecture

Project Structure

`backend/main.py` (FastAPI), `frontend/` (React/Vite), `svelte-frontend/` (Svelte), `scraper/` (Playwright)

Technology Stack

Layer	Technology	Purpose
Backend Framework	FastAPI	REST/API server
Database	PostgreSQL	Data persistence
Auth	JWT	Authentication & authorization
Frontend	React	User interface
Deployment	Render.com + GitHub Pages	Hosting & CI/CD

Module Breakdown

See project structure above.

Data Flow

1. Request enters via HTTP (REST, GraphQL, or WebSocket)
2. Authentication middleware validates JWT token (if required)
3. Router dispatches to appropriate handler
4. Handler processes business logic with database queries
5. Response formatted and returned to client
6. Frontend renders response in appropriate view

Entry Points

- **Server:** `uvicorn backend.main:app`
- **Port:** 8000 (default)
- **Health check:** `GET /health`

Configuration

Environment Variables

Variable	Description	Default
DATABASE_URL	Database connection string	sqlite:///data.db
SECRET_KEY	JWT signing secret	(required)
CORS_ORIGINS	Allowed CORS origins	*
DEBUG	Debug mode flag	false

Deployment

Requirements

- Python 3.10+
- pip dependencies (see requirements.txt)
- SQLite (or PostgreSQL for production)

Startup Commands

```
# Development
uvicorn backend.main:app --reload

# Production
uvicorn backend.main:app --host 0.0.0.0 --port 8000
```

Related Reports

- Report 02: Architecture Analysis (detailed component diagram)
- Report 05: Tech Stack & Dependencies (version-specific)
- Report 13: Deployment Guide (Docker, CI/CD)